

Fit Analysis - Frontend Integration Guide

BecauseFuture | April 2026

1. Fit Classification System

Status	Difference	Color	Meaning
Body Fit	< 0.1 cm	Green	Perfect match
Good Fit	0.1 - 1.9 cm	Light Green	Comfortable
Loose Fit	2.0 - 3.9 cm	Orange	Slightly oversized
Tight	>= 4.0 cm	Red	Too small

2. Screen → API Mapping

Screen	APIs to Call	Auth
Home / Try-On	GET /api/body-measurements (check exists)	JWT
Fit Analysis	1. GET /api/body-measurements 2. GET /api/sizing/garment?url=... 3. POST /api/fit-analysis (each size)	JWT No JWT
Measurements	POST /api/body-measurements	JWT

3. API Reference

3.1 Get Body Measurements

GET /api/body-measurements | Auth: JWT

3.2 Get Garment Data (Brand Portal)

GET /api/sizing/garment?url={product_url} | Auth: None

Returns: measurements_by_size object with sizes as keys (S, M, L, XL...)

3.3 Analyze Fit

POST /api/fit-analysis | Auth: JWT

Body: { "garment_type": "upper|lower", "garment_size": "M", "garment_measurements": {...} }

4. Measurement Mappings

Upper Garments (tops, shirts):

Garment Field	Body Measurement	Transform
breast_width	breast_circumference	÷ 2

front_length	collarbone_to_belly_button_length	direct
arm_length	arm_length	direct
arm_width	biceps_circumference	÷ 2

Lower Garments (pants, jeans):

Garment Field	Body Measurement	Transform
waist	waist_circumference	÷ 2
hip	hip_circumference	÷ 2
front_crotch	waist_to_crotch_front_length	direct
inner_leg_length	inner_leg_length	direct
thigh	upper_thigh_circumference	÷ 2

5. UI Elements → Response Data

UI Element	Data Source
"Your Size: 38"	Size with highest accuracy
"Accuracy: 80%"	$(\text{body_fit_count} + \text{good_fits}) / \text{total} \times 100$
Colored lines on body	measurements[].fit_status → color
Size arrows (< 36 >)	garment.measurements_by_size keys
Garment name	garment.name
Brand	garment.brand_partners.brand_name

6. Color Coding (CSS)

```
--fit-body: #22c55e; --fit-good: #84cc16; --fit-loose: #f97316; --fit-tight: #ef4444;
```

7. Complete API Flow

1. User taps "Fit Info" button
2. GET /api/body-measurements → user measurements
3. GET /api/sizing/garment?url=... → garment with measurements_by_size
4. FOR EACH size: POST /api/fit-analysis → fit result
5. Find best size (highest accuracy)
6. Render UI: body silhouette + colored lines + size panels

8. Accuracy Calculation

```
accuracy = (body_fit_count + good_fits) / (body_fit_count + good_fits + loose_fit_count + tight_count)
× 100
```